



C23-EC-ECII-303

23131

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DECE – THIRD SEMESTER EXAMINATION

DIGITAL ELECTRONICS

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Subtract $(010110)_2$ from $(110110)_2$ using 2's complement method.
2. Realize OR gate using NAND gates.
3. Classify logic families.
4. Define the term fan in and fan out in digital ICs.
5. State the need for tri-state buffer.
6. Draw the logic diagram of 4×1 multiplexer.
7. What is meant by race around condition in JK flip-flops?
8. State the necessity of registers and classify registers based on data I/O operations.
9. Classify different types of semiconductor memories.
10. List the differences between flash ROM and NVRAM.

PART—B

10×5=50

Instructions : (1) Answer *any five* questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. Write the truth table for 2-input NOR gate and realize AND, OR, NOT gates using NOR gates.

12. Simplify $Y = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$ using K-Map.

13. Explain the working of CMOS NAND gate with logic circuit diagram.

14. Explain the working of 4-bit 2's complement adder/ subtractor circuit.

15. Explain the working of 3×8 decoder with logic circuit and truth table.

16. Explain master slave JK flip-flop with necessary diagrams.

17. Explain the working of 4-bit shift right register with circuit diagram.

18. Explain the working of diode ROM with suitable circuit diagram.

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